Replication of a Research Claim from Ma-Kellams et al. (2011),
from *Psychological Science*

Replication Team: Thomas Talhelm

Research Scientist: Melissa Kline

Action Editor: Rich Lucas

Independent Reviewers
(add name below when you initiate review, comment “DONE” on your name when you finish):

Reviewer #1: [Ilka Gleibs

Reviewer #2: [Scott Braithwaite]

Reviewer #3: [NAME]

Review Period: July 29 - August 3

View-only links to: Original Paper, Replication Materials

Replication of a Research Claim from Ma-Kellams et al. (2011), from Psychological Science

SCORE report Ma-Kellams_PsychologSci_2011_lkBL_618

Sourcing notes for this preregistration are here.

Welcome to the replication/reproduction team! You can get started with your preregistration by clicking here.

Privacy Statement: Other teams are making predictions about the outcomes of many different studies, not knowing which studies have been selected for replication. As a consequence, the success of this project requires full confidentiality of the research process, including peer review. This includes privacy about which studies have been selected for replication and all aspects of the discussion about these replication designs.

Table of contents

Table of contents

Sourcing Notes

Instructions for Preregistration Reviewers

Instructions for Replication/Reproduction Team

Preregistration for SCORE

- Study Information
  - 1. Title (provided by SCORE)
  - 2. Authors and affiliations

- 3. Description of study (provided by SCORE).
- 4. Hypotheses
- Design Plan
  - 5. Study type (provided by SCORE)
  - 6. Blinding (multiple choice question)
  - 7. Blinding (free response)
  - 8. Study Design
  - 9. Randomization
- Sampling Plan
  - 10. Existing data (multiple choice question, provided by SCORE)
  - 11. Explanation of existing data (provided by SCORE)
  - 12. Data collection procedures
  - 13. Sample size
  - 14. Sample size rationale
  - 15. Stopping rule
- Variables
  - 16. Manipulated variables
  - 17. Measured variables
  - 18. Indices
- Analysis Plan
  - 19. Statistical models
  - 20. Transformations
  - 21. Inference criteria
  - 22. Data exclusion
  - 23. Missing data
  - 24. Exploratory analysis
  - 25. Other
- Final reviewer checklist

Bibliography

Sourcing Notes

The research scientists write a short set of constraints or notes about each study that are designed to help match the replication/reproduction to a team that will perform it. They are maintained as part of the project record.

- SCORE Research Scientists insert sourcing notes for Chris here.

Instructions for Preregistration Reviewers

Your role is to review preregistered research designs for clarity, completeness, and quality. For the purposes of a SCORE replication, a preregistration is high quality if it generates a protocol that is a good faith attempt to replicate the original finding. In sum, focus on whether differences in original versus replication protocols are substantively anticipated to matter for claims in the original paper (and the broader field), and be biased against spending time on speculative concerns that do not have an evidence base.

Please start by reading the Reviewer Criteria checklist. For each section, you should evaluate whether the description is complete, whether deviations from the original study (or additions, if the information was not available) are documented, and whether, all told, the decisions are consistent with a good faith replication. You don't need to fill out a copy of the checklist, but should use it as a guide to the types of information that should be present in a finished preregistration - remember that not all items will apply to all projects. At the very end of the pregistration is a final reviewer checklist where you can give your evaluation of the preregistration as a whole. When you are completely finished reviewing, please comment 'DONE' on your name on the first page of this document.

These replications are intended to be robust, high quality studies, and in some cases, this will involve deviation from the original study. For instance, all SCORE projects are preregistered, and all use sample sizes that are based on a formal power analysis, whether or not these were the case in the original study. Labs may choose to include additional 'best practices' that may not have been present in the original study, in addition to other necessary differences between the original and replication study. These are allowable, so long as they remain a good faith replication of the original finding. We are collecting a list of best practices which are the kinds of steps that labs are encouraged to take (and that you can recommend!) to increase the robustness and replicability/reproducibility of their work. Keep in mind that not all projects need to (or can) include all of these practices.

The preregistration for this paper begins here. You can also reference this list of Frequently Asked Questions for preregistration reviewers.

Privacy Statement: Other teams are making predictions about the outcomes of many different studies, not knowing which studies have been selected for replication. As a consequence, the success of this project requires full confidentiality of the research process, including peer review. This includes privacy about which studies have been selected for replication and all aspects of the discussion about these replication designs.

Instructions for Replication/Reproduction Team

General information about preregistration is available at https://cos.io/prereg. Every section should have a response from you; in the case that a section truly does not apply (e.g., "Manipulated Variables" do not exist in an observational study), you can respond with "N/A". Some sections are indicated as multiple choice, and we ask that you bold your response(s) if one has not already been selected. All other sections are open-ended.

A Research Scientist from the Center for Open Science has provided foundational information for your SCORE project from the original paper and, where possible, additional feedback and materials from the original author(s). **This information should not be considered a complete response to a section unless otherwise noted.**

The preregistration for your replication or reproduction should provide as detailed a plan as possible of what **you** will be doing, not just describe what was done in the original paper. That plan should be written in the future tense, and reference the original paper and any original materials or correspondence with the original author as necessary to provide context or justification of any decisions made for your protocol.

You are encouraged to look over the Reviewer Criteria that will be used to evaluate your preregistration. For each question, your response should include a complete **description** (state what you will do, in enough detail that others could implement your plan), list of **deviations** (clearly state anything you added, omitted, or changed from the original study), and **rationale** (justify why your decisions are consistent with a good-faith replication of the original claim).

Materials developed for a replication do not need to be included directly within the preregistration document. Instead, please upload any materials (such as surveys, audio/visual stimuli, instructions for coders/confederates, etc.) to the "Methods and Materials" component of the OSF project for your SCORE protocol. If you create a codebook/data dictionary or dummy dataset for review, please upload these to the "Data" component. You should also upload your data cleaning and data analysis scripts to the "Analysis" component. Although we encourage you to prepare these scripts in time for the external review of your preregistration, you are not required to do so. However, you will have to provide such a script before beginning data collection for SCORE, so we strongly encourage developing it as early as possible. Any files you upload for your project should be directly referenced within the document by filename so it is clear what is being used for your replication and where reviewers can find it. (For instance: *"Each participant will see 16 of the 64 cat pictures. All stimuli are uploaded to the Methods & Materials component, as cat1.jpg, cat2.jpg...cat64.jpg."*)

You can reach out to the SCORE Project Coordinators for additional guidance at scorecoordinator@cos.io, and you can also reference this list of Frequently Asked Questions.

Preregistration for SCORE

Study Information

1. Title (provided by SCORE)

RR TEAM INSTRUCTIONS: *This has been determined by SCORE.*

Direct Replication of a research claim from Ma-Kellams & Blascovich, 2011, in Psychological Science

2. Authors and affiliations

RR TEAM INSTRUCTIONS: *Fill in the names and affiliations of your team below.*

Thomas Talhelm[1]

1 University of Chicago Booth School of Business 1

3. Description of study (provided by SCORE).

RR TEAM INSTRUCTIONS: *This description has been provided by SCORE. Please review and make a SCORE project coordinator aware of any edits, additions, and corrections you would suggest to the paragraph. You are free to add additional descriptions of your project in a separate paragraph.*

The claim selected for replication from Ma-Kellams & Blascovich, 2011 is that after priming of mortality salience (vs. a control prime for dental pain), European Americans become derogating of dissimilar others (i.e. expressing negative attitudes toward prostitution after reading about a prostitution defendant). In contrast, East Asian participants' approval of prostitution increased after priming of mortality salience (compared to control prime). The explanation is that, in the control condition, Asian Americans tend to be relatively socially conservative, but under mortality-salience conditions, they magnify the general East Asian tendency toward collectivism and interdependence to the point of embracing non-normative individuals. This reflects the following from the paper's abstract: "Given fundamental East-West differences in self-construal (i.e., the independent vs. interdependent self), we predicted that members of Eastern cultural groups would affirm other people, rather than defend and affirm the self, after encountering

conditions of mortality salience." The main analysis was a 2 (condition: mortality salience vs. control priming task) × 2 (ethnicity: European American vs. Asian American) univariate ANOVAs on the dependent variable of the Prostitution Attitudes Scale scores, which yielded a significant interactive effect on PAS (Prostitution Attitudes Scale) score, $F(1, 38) = 10.263$, $p = .003$

4. Hypotheses (provided by SCORE with possible RR team additions)

RR TEAM INSTRUCTIONS: *The focal test for SCORE is indicated as H*. If you will test additional hypotheses (or use alternate analyses) that help you to evaluate the claim your replication/reproduction is testing, number them H1, H2, H3 etc. (You can place H* in the list wherever makes sense). Please make sure that any additional hypotheses are logical deductions/operationalizations of the selected SCORE claim or are necessary to properly interpret the focal H* hypothesis. Research that is outside this scope should be described in a separate preregistration.*

Specific points to keep in mind (please also consult the [Reviewer Criteria]):

- *Are the listed hypotheses specific, concise, clearly testable, and specified at the level of operationalized variables?*
- *Are hypotheses identified as directional or non-directional, and, if applicable, have the direction of hypotheses been stated? (Example: "Customers' mean choice satisfaction will be higher in the CvSS architecture condition than in the standard attribute-by-attribute architecture condition.")*
- *Does the list of hypotheses/tests indicate whether additional hypotheses are taken from the original study or modified/added by the team?*

H* (SCORE focal test): That there will be an interaction effect between ethnicity (European American vs. Asian American) and priming condition (mortality salience vs. pain control task) on participants' Prostitution Attitudes Scale scores, such that after mortality salience, Asian American participants will show more positive attitudes toward prostitution and European American participants will show less positive attitudes toward prostitution in the mortality salience condition than in the control condition.

[CONTINUE QUESTION 4 RESPONSE HERE]

Design Plan

5. Study type (multiple choice question)

NOTE: *Select one option from the list below that applies for your study by bolding it.*

- **Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.**
- Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, natural experiments, and regression discontinuity designs.
- Meta-Analysis - A systematic review of published studies.
- Other

6. Blinding (multiple choice question)

RR TEAM INSTRUCTIONS: *Select any/all of the below that apply for your study by bolding them. You will give a longer description in the next question.*

- No blinding is involved in this study.
- **For studies that involve human subjects, they will not know the treatment group to which they have been assigned.**
- **Personnel who interact directly with the study subjects (either human or non-human subjects) will not be aware of the assigned treatments. (Commonly known as "double blind")**
- Personnel who analyze the data collected from the study are not aware of the treatment applied to any given group.

[QUESTION 6 - BOLD YOUR RESPONSE ABOVE]

7. Blinding (free response)

RR TEAM INSTRUCTIONS: *Please describe the blinding procedures for your study here, including enough detail to allow the reviewers to evaluate your plan. If the details of a blinding procedure are closely tied to the experimental protocol, you can refer to longer descriptions (e.g. in your Data Collection response) so long as the information is available somewhere in the preregistration.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration comment on blinding of both participants and study personnel?*

- *Does the preregistration comment on blinding of both hypotheses and condition assignment?*
- *If the original materials do not provide substantial detail on the blinding procedures, is it clear what additions the replication is making?*

Participants will not be told what condition they are assigned to or the hypothesis being tested. However, because the manipulation involves a stimulus that participants will consciously process (dental pain versus mortality salience), participants will have some awareness of the manipulation. Participants may also be able to guess the intention of the experiment.

8. Study Design

RR TEAM INSTRUCTIONS: *In this section, state your study design. Depending on the type of study you are conducting, this may be very brief (i.e. listing the factors and how they are manipulated, such as "2 (Color: Red/Blue) x 2 (Height:Tall/Short), between subjects"), or may be much longer. For instance, observational studies may involve more precise specification of the population and sampling strategy, or discussion of inferences involving assumptions about causal effects. Examples of study design include two-group, factorial, randomized block, and repeated measures. Is it a between (unpaired), within-subject (paired), or mixed design? Typical study designs for observation studies include cohort, cross sectional, and case-control studies.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration specify the unit of analysis?*
- *Does the preregistration describe how many treatment conditions will be used in the study, and how many conditions participants will be exposed to?*
- *Does the preregistration provide sufficient detail about how the study design deviates from or is congruent with the study design employed in the original study?*

The main analysis will be a 2 (condition: mortality salience vs. control priming task) × 2 (ethnicity: European American vs. Asian American) multivariate ANOVA on the dependent variable of the Prostitution Attitudes Scale scores. The original paper found a significant interactive effect on PAS (Prostitution Attitudes Scale) score, $F(1, 38) = 10.263$, $p = .003$

9. Randomization (free response)

RR TEAM INSTRUCTIONS: *If you are doing a randomized study, state how will you randomize, and at what level. If you will not randomize some factors in your study design, please draw a clear distinction regarding which factors are randomized, and how other factors are distributed or determined across units of analysis.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration describe the level at which randomization takes place? (Examples include randomization within subject, blocked by condition, by cluster, etc.)*
- *Does the preregistration describe the method of randomization that is used? (Examples include simple, block, stratified, and adaptive covariate randomization, etc.)*
- *Does the preregistration describe how the randomization is implemented? (Examples include Kish grid, random number table, statistical software package, etc.)*

Participants will be randomized to either (1) writing about dental pain or (2) writing about death. This is a between-participants design, and it will be assigned randomly by Qualtrics. However, the variable of cultural background (European American versus Asian American) is not randomly assigned. To ensure roughly equal cell sizes, we will choose the option in Qualtrics that allows for equal cell sizes (i.e., among the Asian American participants, equal cell sizes in dental pain versus death writing conditions, and among the European American participants, equal cell sizes in dental pain versus death writing conditions.

Sampling Plan

In this section we'll ask you to describe how you plan to collect samples, as well as the number of samples you plan to collect and your rationale for this decision. Please keep in mind that the data described in this section should be the actual data used for analysis, so if you are using a subset of a larger dataset, please describe the subset that will actually be used in your study.

10. Existing data (multiple choice question, provided by SCORE)

- **1.1.1. Registration prior to creation of data**
- 1.1.2. Registration prior to any human observation of the data
- 1.1.3. Registration prior to accessing the data
- 1.1.4. Registration prior to analysis of the data
- 1.1.5. Registration following analysis of the data

11. Explanation of existing data (provided by SCORE)

NOTE: *For a replication, this question refers to the data from the replication itself, not the original study. Even if we have access to the data from the original study, that is* ***not*** *the data that will be used for the replication of the claim and does not need to be included in this question.*

12. Data collection procedures

RR TEAM INSTRUCTIONS: *Please describe the process by which you will collect your data. If you are using human subjects, this should include how you will identify the population from which you obtain subjects, recruitment efforts, payment for participation, how subjects will be selected for eligibility from the initial pool (e.g. inclusion and exclusion rules), and your study timeline, in addition to the experimental/observational protocol itself. For studies that don't include human subjects, include information about how you will collect samples, duration of data gathering efforts, source or location of samples, or batch numbers you will use.*

Where details are described in other questions (e.g. study design, blinding) you can refer to those questions, so long as the complete description is provided somewhere. You are strongly encouraged to supplement your description here with materials (which might include stimuli, survey instruments, code for running data collection software, instructions for experimenters) uploaded to your OSF project, in the "Materials and Methods" component. Please use the specific file names when referencing them in your description.

Specific points to keep in mind (please also consult the [Reviewer Criteria]):

- *Does the preregistration describe the target population, and how members of the target population are sampled for the study?*
- *Does the protocol describe in detail any materials that will be presented to participants, including variation or structure in those materials relevant to the experimental design? Are these materials available for review?*

Participants will be recruited through the University of Chicago Booth School of Business Center for Decision Research participant pool. Because the original study used undergraduate students, we will selectively sample undergraduate students. If sampling undergraduates does not allow us to reach our target sample size, we will expand the sample to include graduate students and MBA students.

Also, because the original study compared European American versus East Asian American students, we will selectively recruit participants from these two cultural backgrounds. The original study included East Asian participants born abroad but now living in the US, as well as East Asian participants born in the US but whose parents were born abroad. Thus, we will follow this design, including both US-born and non-US-born participants of East Asian backgrounds.

The original authors provided the original materials for the (1) experimental manipulation--writing about dental pain or death, (2) the prostitution case description that participants read after the writing manipulation, and (3) the attitudes toward prostitution scale (PAS) and the bail amount.

We decided to make one change from the original materials. The original study took place in Santa Barbara, California, and the prostitution case materials described California. Because our participants will be in Chicago, we determined it would be best to edit these details to be about Chicago. Of course, deviations in the study materials can cause problems. However, research on construal level has found that people respond differently when told about situations that are far away as opposed to nearby (Trope & Liberman, 2010). Thus, it is plausible that asking participants in Chicago to think about California could affect their responses. In order to prevent this, we changed (1) references about California to Illinois, (2) references to Santa Barbara to Chicago, (3) street names from names common in California (e.g., "Cielo Lane") to names of streets in Chicago (Union and Perry Ave.), (4) references from the original county to Cook County (where Chicago is located). For the same reason, we also changed the years recorded in the case to 2019 and 2020.

The original authors did not include the original question wordings for the demographic questions, so we created these anew. We added these to the end of the study (after participants have completed the writing, case, and attitude measures).

Trope, Y., and Liberman, N. (2010). Construal level theory of psychological distance. *Psychological Review* 117: 440. doi:10.1037/a0018963

13. Sample size

RR TEAM INSTRUCTIONS: *The analytic sample sizes below come from the SCORE power analysis (see next question). These sample sizes do not account for participant attrition, data exclusions, or otherwise missing data. Your actual recruited sample will likely need to be larger than these analytic sample sizes in order to attempt to arrive at a sufficiently powered analysis. It is at your discretion to propose a sampling approach and rationale to address this difference in target* ***recruited*** *vs target* ***analytic*** *sample sizes - you can use the suggested language below or frame it in your own words.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the description of the analytic sample size (and target sample size for recruitment, if this differs) include targets for both first round and second round data collection?*
- *If there is more than one possible 'sample size' that could be referred to (cell vs. total size, cluster levels, whole design vs. subset for focal analysis), is the distinction made clear?*

The initial target analytic sample size is **72 participants**. If a statistically significant effect is not observed after the first round of data collection, a second round will begin. The second round of data collection will sample an additional **86 participants** for a target pooled analytic sample of **158 participants.** We will attempt to recruit an equal number of participants per each cultural group.

[CONTINUE QUESTION 13 RESPONSE HERE]

IF APPLICABLE: To achieve the target analytic sample, we anticipate the need to collect a larger target sample size to account for _____. The target sample size for recruitment is _____ [units of observation]. If necessary, the second round of data collection will sample an additional _____ [units of observation] for a pooled recruited sample size of _____ [units of observation].

14. Sample size rationale

NOTE: *Power calculations for SCORE protocols are performed by either a Research Scientist at the Center for Open Science or by one of our consultants. In some cases, the power calculation will not yet be done for your protocol by the time you begin work on it; if you urgently require a defined sample size in order to submit your IRB application or otherwise make progress on your protocol, please contact the Project Coordinators (scorecoordinator@cos.io) so we can prioritize your protocol. Otherwise, please be patient as we complete these calculations and we will notify you when the target sample size has been defined.*

Power calculations were done in accordance with the guidelines of the Social Sciences Replication Project (SSRP). The first round of data collection achieves 90% power to detect 75% of the original effect size. The pooled sample, if necessary after testing the effect on the first round of data collection, achieves 90% power to detect 50% of the original effect size.

The power analysis for this project was conducted by Melissa Kline, and can be found at this osf link.

15. Stopping rule

RR TEAM INSTRUCTIONS: *The first paragraph of this response refers to the two-stage data collection strategy for SCORE. Beyond this, your data collection procedures may not give you full control over your exact sample size; specify here how you will decide when to terminate your data collection. You will describe the specifics of what data is excluded in question 22, but please describe here how you will determine when to stop collecting new data, aiming to meet your analytic sample size.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *If the stopping rule is based on an estimated fall-off rate (e.g. attrition rate), is that rate justified? Does the plan specify how you will proceed if the resulting sample size is somewhat over or under the target?*
- *If the study includes post-collection data exclusion (e.g. participants are excluded who fail manipulation checks; analytic sample consists of all who fall below 1SD on some measure), does the stopping rule allow you to track inclusion without seeing the critical results?*
- *Does the preregistration make clear whether the plan for finishing data collection follows or deviates from the original study? If the original study is silent on an explicit stopping rule, is this made clear?*

The planned sample size is **72**. After achieving that sample, planned analyses will be run. If a significant effect in the hypothesized direction is found, sampling stops. If that significant effect

is not found, a second round of data collection will collect data from **86** additional observations, for a pooled sample of **158**. Sampling will stop after the second round of data collection regardless of a significant effect.

[CONTINUE QUESTION 15 RESPONSE HERE]

In order to achieve the necessary sample size, we will recruit from the Center for Decision Research participant pool. We will first attempt to recruit undergraduate students because the original study tested undergraduate students. However, if we fail to achieve the desired sample size with undergraduate students, we will open up recruitment to graduate students and MBA students in the pool.

Variables

RR TEAM INSTRUCTIONS: *The preregistration form divides variables across three questions: manipulated variables, measured variables, and indices (i.e. analytic variables derived from raw variables). Transformed variables (e.g. reaction time → log reaction time) can be defined here as well; you will discuss how those transformations are calculated in the analysis section.*

Across these questions, you should define all variables that will later be used during your analysis (including data preparation/processing). You can describe all variables in the preregistration and/or summarize and link to a data dictionary (codebook) in your repository to answer these questions; please make sure to indicate which variables are manipulated and which are measured.

If you will share data from your replication, this is also the place to state whether any variables will be removed prior to sharing the dataset (e.g. to reduce risk of participant identification or comply with copyright restrictions on scale items.)

16. Manipulated variables

RR TEAM INSTRUCTIONS: *Describe all variables you plan to manipulate and the levels or treatment arms of each variable (not applicable to an observational study). For any experimental manipulation, you should give a precise definition of each manipulated variable, e.g. "loud or quiet," should instead give either a precise decibel level or a means of recreating each level. 'Presence/absence' or 'positive/negative' is an acceptable description if the variable is precisely described. You can also refer to your data collection protocol if variable levels are more completely described there.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *For each of these variables, does the preregistration describe how each condition will be manipulated?*
- *Does the preregistration comment on the role of each manipulated variable in the focal analyses (e.g., independent variable, moderator, etc)?*
- *Does the preregistration describe any changes from the original study in procedure, context, or instruments used for these manipulated variables (e.g., sound condition played over headphones instead of speakers)?*

The key experimental manipulation is having participants write about either dental pain or about death. The original researcher provided this manipulation and the dependent variables to the replicators. The other contrast is between East Asian Americans versus European Americans. This cultural background is, of course, not experimentally manipulated. These demographic questions were not provided to the replication team.

We will ask about age, gender, cultural background of participants and each parent, and student status (undergrad, grad student, MBA).

17. Measured variables

RR TEAM INSTRUCTIONS: *Describe each variable you will measure, including outcome measures, as well as any predictors, covariates, or descriptive information that you will measure. As with the previous questions, the answers here must be precise. For example, 'intelligence,' 'accuracy,' 'aggression,' and 'color' are too vague. Acceptable alternatives could be 'IQ as measured by Wechsler Adult Intelligence Scale', 'percent correct,' 'number of threat displays,' and 'percent reflectance at 400 nm.'*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration comment on the role of each measured variable in the focal analyses (e.g., inclusion criteria, dependent variables, control variables, etc)?*
- *If the study will measure variables which will not be involved in the focal analysis, are these variables disclosed?*
- *Does the preregistration describe any changes in procedure, context, or instruments used for these measured variables (e.g., extraversion measured with EPI vs MIES)?*

The main dependent variable is (as described in the original study): the Prostitution Attitudes Scale (PAS; Rosenblatt et al., 1989), which consists of five items about prostitution (two against prostitution and three favoring prostitution); ratings are made on Likert scales ranging from −3 (*strongly disagree*) to +3 (*strongly agree*). The two antiprostitution items will be reverse-coded (following original study), and ratings for the five items were averaged to yield an overall prostitution attitude score, with higher scores reflecting more favorable attitudes toward prostitution.

Rosenblatt and colleagues (1989) performed a factor analysis on this scale and found that the "factor analysis of this scale revealed only one factor, with all five items loading .70 or higher." The Ma-Kellams study did not report a factor analysis but found that the scale evinced satisfactory internal consistency, with a Chrobach's alpha of 0.74. It is the replicator's judgment that the items show simple face validity.

The original study also asked participants to set a bail amount for prostitution. However, this was not the main dependent variable. We will present these analyses in the report, but they are not the criterion of replication.

18. Indices

RR TEAM INSTRUCTIONS: *If any of the measured variables described in Section 17 are going to be combined into a composite measure (including simply a mean), describe what measures you will use and how they will be combined. Include either a formula or a precise description of your method. If you are using a more complicated statistical method to combine measures (e.g. a factor analysis), you can note that here but describe the exact method in the analysis plan section.*

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration specify each of the composite measures (e.g. mean scores, factor scores) that are needed for the focal analysis, and which of the measured variables in Section 17 are used in each one (e.g. the happiness, joy, and satisfaction items will be used to create the 'positive feelings' measure)?*
- *Does the preregistration provide a detailed description of the methodology or a precise formula that will be used to construct each composite measure?*

To calculate scores on the Prostitution Attitudes Scale, we will follow the original study: The Prostitution Attitudes Scale (PAS; Rosenblatt et al., 1989), which consists of five items about prostitution (two against prostitution and three favoring prostitution); ratings are made on Likert scales ranging from −3 (*strongly disagree*) to +3 (*strongly agree*). The two antiprostitution items were reverse-coded, and ratings for the five items were averaged to yield an overall prostitution attitude score, with higher scores reflecting more favorable attitudes toward prostitution.

Analysis Plan

19. Statistical models

RR TEAM INSTRUCTIONS: *This section should describe in detail the analysis that will be performed to replicate the focal result. This analysis must align as closely as possible with the original study's analysis, even if you have identified limitations in the original study. The level of detail should allow anyone to reproduce your analyses from your description below. Examples of what should be specified: the model; each variable; adjustments made to the standard errors and to case weighting; additional analyses that are required to set up the focal analysis ; the software used.*

Beyond the replication of the focal analysis from the original study, it is at your discretion to test the claim using other analytic approaches as a check of the robustness of the claim. The original test should be listed first and be clearly distinguished from any other tests. If you are testing additional confirmatory hypotheses, describe them in the same order as you numbered

them in the "Hypotheses" section above and make clear reference to the specific hypothesis being tested for each.

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration specify which statistical model will be used to provide the 'focal evidence' for the SCORE test (e.g. a regression coefficient in a larger multiple regression model), and does it correspond closely to the model and evidence from the original study?*
- *Does the preregistration describe each variable that will be included in the focal analysis, and what role each variable has (e.g. dependent variable, independent variable)?*
- *Does the preregistration include a detailed specification of the focal analysis, including interactions, lagged terms, controls, etc.?*

For the purposes of SCORE, to test **H*** in line with the original paper, [I/we] will use...

The main analysis will be a 2 (condition: mortality salience vs. control priming task) × 2 (ethnicity: European American vs. Asian American) multivariate ANOVA on the dependent variable of the Prostitution Attitudes Scale scores. The original paper found a significant interactive effect on PAS (Prostitution Attitudes Scale) score, $F(1, 38) = 10.263$, $p = .003$.

For data checks, we will calculate a Chronbach's alpha for the PAS. The original study found a Chronbach's alpha of 0.74. We will report the Chronbach's alpha for the PAS in our sample and report the alpha from the published paper so that readers can compare the two numbers.

There are different opinions on what alpha values are unacceptable, but one source suggests values under 0.50 are generally unacceptable. Thus, we will consider an alpha of under 0.50 to have unacceptable consistency. Whatever the number, we will report this statistic in the paper so that readers will be aware of it. However, we will continue to run the statistical interaction test.

To check for outliers, we will produce a histogram of the PAS scores in SPSS. We will check for outliers of more than 2 standard deviations on PAS total scores. To follow the original, we will run the main statistical interaction test with all participants. However, as a robustness check, we will also present the interaction test with > 2 SD outliers removed (if any).

Another critical question is the direction of the interaction. Strictly speaking, a significant interaction would fit the criterion of a successful replication. However, researchers will also be interested in whether the direction of the interaction fits the direction of the original study. To break this out, we will (1) produce a figure akin to Figure 1 in the

original article (culture x writing condition on the DV of PAS score) and (2) planned statistical comparisons reporting means and t tests. The planned comparisons in the original article were reported as follows:

> **Planned comparisons showed results consistent with existing TMT research: European Americans reported a significantly more negative attitude toward prostitution in the mortality-salience condition (M = −0.0092, SD = 0.28) than in the control condition (M = 1.022, SD = 0.34), t(20)= 2.10, p = .048, whereas Asian Americans showed the opposite effect (mortality-salience condition: M = 0.21, SD = 0.31; control condition: M = −0.73, SD = 0.34), t(18) = 2.73, p = .013).**

In other words, for European American participants, mortality salience decreased PAS scores compared to the dental pain condition. For East Asian participants, mortality salience increased PAS scores compared to the dental pain condition.

20. Transformations

RR TEAM INSTRUCTIONS: *This section should describe how any of the measured variables or composite measures mentioned above will be transformed prior to the analyses listed in Section 19. These are adjustments made to variables* ***after*** *measurement or measure creation, and might include centering, logging, lagging, rescaling etc. Please provide enough detail such that anyone else could reproduce the transformations based on the description below.*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration specify which of the measured variables or composite measures will need to be transformed prior to the focal analysis?*
- *For each variable needing transformation, does the preregistration adequately describe the transformations, including any centering, logging, lagging, recoding, or implementation of a coding scheme for categorical variables?*

- *For any categorical predictors that are included in a regression, does the preregistration indicate how those variables will be coded (e.g. dummy coding, summation coding, etc.) and what the reference category will be?*

No transformations are planned.

21. Inference criteria

RR TEAM INSTRUCTIONS: *This section describes the precise criteria that will be used to assess whether the hypotheses listed above were confirmed by the analyses in Section 19. The default language below only applies to the test of the SCORE claim, **H*.** It is at your discretion to describe the inferential criteria you will use for any additional analyses. They need not rely on p-values and/or the same alpha level we have specified for **H*.***

If the additional analyses will use multiple comparisons, the inference criteria is a question with few "wrong" answers. In other words, transparency is more important than any specific method of controlling the false discovery rate or false error rate. One may state an intention to report all tests conducted or one may conduct a specific correction procedure; either strategy is acceptable.

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *For each hypothesis listed in the Hypotheses section above, does the preregistration clearly describe the inference criteria necessary to call the replication attempt for that hypothesis test successful, specified at the level of the operationalized variables?*
- *Example:"...For this replication attempt, this criteria is met by a statistically significant difference in customers' mean choice satisfaction in the CvSS architecture condition compared to the standard attribute-by-attribute architecture condition, with the mean higher in the CvSS architecture condition than in the standard attribute-by-attribute architecture condition."*

Criteria for a successful replication attempt for the SCORE project is a statistically significant effect (alpha = .05, two tailed) in the same pattern as the original study. For the SCORE focal test (H*), this criteria is met by an interaction effect between ethnicity (European American vs. Asian American) and priming condition (mortality salience vs. pain control task) on participants' Prostitution Attitudes Scale scores, such that after mortality salience, Asian American participants will show more positive attitudes toward prostitution. Meanwhile, European American participants will show less positive attitudes toward prostitution in the mortality salience condition than in the control condition.

Because multiple studies in terror management have found that mortality salience increases harsh judgment and punishment of deviators (McGregor et al., 1998; Florian & Mikulincer, 1997; Milkulincer & Florian, 2000; Rosenblatt et al., 1989), it is reasonable to expect to reproduce the

direction of the main effect of mortality salience increasing disapproval among the European American participants. However, there is less cross-cultural research testing the effect among East Asian and East Asian American participants. One could expect to find a significant main effect in the opposite direction (mortality salience decreasing disapproval) among East Asian participants because the original study found this pattern. However, one could also argue that the lack of a main effect in a different culture is still a valid and worthwhile cultural difference. For example, research on cognitive dissonance across cultures has found that it simply doesn't work in samples from some cultures (or only under particular conditions not present in the original dissonance studies: Hoshino-Browne et al., 2005). Thus, we think either pattern among the East Asian participants would be consistent with the demonstration of a cultural difference in line with the earlier study.

[CONTINUE QUESTION 21 RESPONSE HERE]

22. Data exclusion

RR TEAM INSTRUCTIONS: *The section below should describe the rules you will follow to exclude collected cases from the analyses described in Section 19. Note that this refers to exclusions **after** data collection; exclusion criteria that prevent a case from entering your recruitment sample should be described in earlier sections. Please be as detailed as possible in describing the rules you will follow (e.g. What is the specific definition of outliers you will use? Exactly how many attention checks does a participant need to fail before their removal from the analytic sample?).*

Specific points to keep in mind (please also consult the Reviewer Criteria):

- *Does the preregistration comment on whether any cases recruited for the study sample will be excluded prior to data analysis?*
- *If yes, does the preregistration provided detailed instructions on how the exclusions will be performed (e.g. Is the definition of outlier provided? Is the number of attention checks failed before a participant is excluded specified?)*

Participants of European American and East Asian (Japan, Korea, Mainland China, Hong Kong, Taiwan, Singapore) American backgrounds will be recruited. We will follow the original study and include East Asian participants born abroad and those whose parents were born abroad. Participants will be excluded from analysis if they do not report belonging to one of these groups or if they report a mixed cultural background. No attention checks will be conducted.

The original study did not report any checks of the writing manipulation (for example, if the participant does not write anything or writes very little in the manipulation). Thus, we will present the main analysis with all participants. However, we will also present exploratory analyses (1) excluding any participant who wrote nothing and (2) excluding participants who wrote 1 sentence or fewer.

23. Missing data

RR TEAM INSTRUCTIONS: *The section below should describe how missing or incomplete data will be handled. Please be as detailed as possible in describing the exact procedures you will follow (e.g. last value carried forward; mean imputation) and any software required (e.g. We will use Amelia II in R to perform the imputation).*

Specific points to keep in mind (please also consult the [Reviewer Criteria]):

- *Does the preregistration comment on how missing or incomplete data will be addressed (e.g. casewise removal, missing data imputation)?*
- *If applicable, does the preregistration specify how many missing variables will lead to a case's removal (e.g. If a subject does not complete any of the three indices of tastiness, that subject will not be included in the analysis.)?*
- *If applicable, does the preregistration describe how missing data imputation will be performed, including relevant software?*

We do not plan on imputing missing data. For the critical test of replication (culture x experimental condition on the DV of PAS attitudes), data is required on the following variables:

- Cultural background
- Experimental condition (and at least some writing for the experimental prime)
- PAS scale

Because all of these variables are required for the statistical test, participants will be automatically excluded from the analysis if they are missing data on any of these variables (including individual questions on the multi-item PAS scale). We do not plan to exclude participants with missing data for other variables, except for when those variables are present in an analysis (for example, when analyzing responses to the bail question, participants with missing data for the bail question will automatically be excluded).
[QUESTION 23 RESPONSE HERE]

24. Exploratory analysis (Optional)

RR TEAM INSTRUCTIONS: *If you plan to explore your data set to look for unexpected differences or relationships, you may describe those tests here. An exploratory test is any test where a prediction is not made up front, or there are multiple possible tests that you are going to use. A statistically significant finding in an exploratory test is a great way to form a new confirmatory hypothesis, which could be registered at a later time. If any exploratory analyses involve additions to the data collection procedure beyond what was performed in the original study (e.g. additional items on the survey; running another condition in the experiment), please describe them below.*

We will run the other analyses reported in the paper that are not the main analysis. Those are:

- **Culture x Condition on amount of bail**
- **T-test comparing cultures in control condition for DV of PAS score**
- **T-test comparing cultures in experimental condition for DV of PAS score**
- **T-test comparing cultures in control condition for DV of bail**
- **T-test comparing cultures in experimental condition for DV of bail**
- **To test the possibility that mood might drive the effects, we will calculate net positive-mood scores using the PANAS (i.e., by subtracting the ratings given to the negative-mood items from the ratings given to the positive-mood items). Test: Main effect of mood score on PAS DV.**
- **Interaction between mood scores and condition on PAS DV.**
- **Main effect of mood score on bail DV.**
- **Interaction between mood scores and condition on PAS DV.**

We will explore potential demographic differences:

- **Religion: we will analyze the effect of religion (all religions versus agnostic/atheists) on the PAS DV. It is plausible that participants from religious backgrounds will have harsher views toward prostitution.**
- **Religion: we will analyze the effect of religion (all religions versus agnostic/atheists) on the bail DV. It is plausible that participants from religious backgrounds will have harsher views toward prostitution.**
- **We will re-run the main interaction (culture x condition on PAS DV) with foreign-born East Asian participants only to test whether the effect size is stronger for people born abroad. It is plausible that East Asian participants born in the US will display smaller cultural differences than East Asian participants born abroad.**
- **Purpose: These demographic differences are meant to be exploratory. They may be useful to understand whether the demographic differences (1) explain variation in the outcome and (2) might be useful moderators for future researchers looking to replicate or extend this effect.**

25. Other

RR TEAM INSTRUCTIONS: *This section serves two purposes. First, please use this section to discuss any features of your replication plan that are not discussed elsewhere. Literature cited, disclosures of any related work such as replications or work that uses the same data, plans to make your data and materials public, or other context that will be helpful for future readers would be appropriate here. Second, please also re-surface any major deviations from earlier in the preregistration that you expect a reasonable reviewer could flag for concern. Give a*

summary of these deviations, focusing on larger changes and any possible challenges for comparing the results of the original and replication study.

Specific points to keep in mind (please also consult the [Reviewer Criteria](#)):

- *Does the preregistration reference other sections of the preregistration where substantial deviations from the original study have been described (including deviations due to differences in location or time compared to the original study)?*
- *Does the preregistration comment on plans to make the data and materials from the replication study public?*

The original paper did not mention whether the study was conducted online or in the laboratory. However, given the data of publication (2011) and the age of participants (20-22 years old), it's likely that participants (1) were college students and (2) participated in the lab as a part of a participant pool. We plan to conduct the study online, which we presume is a difference from the original study. We anticipate that this difference does not have a high likelihood of affecting the results, but it is worth pointing out this potential for discrepancy.

Another difference worth noting is that we plan on conducting the study in the summer and fall of 2020, during the coronavirus pandemic. It's possible that writing about death carries different psychological meanings at a time of widespread disease and fear.

Final review checklist

REVIEWER INSTRUCTIONS: *For the following questions, reviewers please indicate whether you can 'sign off' on the following items by adding a comment. You can update this response as the lab moves through revisions during the review period!*

- Included with this pre-registration are the specific materials needed to conduct the replication (surveys, stimuli, etc). If not,
    - Have the pre-registration authors detailed when these materials will be made available prior to final registration?
    - Can you evaluate whether this preregistration represents a good-faith replication of the original study without seeing these materials?
- Included with this pre-registration are the specific analytic scripts/code/syntax that will be used for the final analysis. If not,
    - Have the pre-registration authors detailed when these analyses will be made available prior to final registration?
    - Can you evaluate whether this preregistration represents a good-faith replication of the original study without seeing these materials?
- I have reviewed all sections of this pre-registration, and I believe it represents a good-faith replication attempt of the original focal claim.